Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/09/2022
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	11/04/2021	**End Date:**	11/03/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/03/2023 per NSSC information (Ed., 11/3/22)		
Key Personnel Changes/Previous PI:	This is a transfer of funding from Dr. Elison Blancaflor (formerly at the Noble Research institute) to the University of Wisconsin-Madison. The postdoc working on the project (Dr. Sabrina Chin) moved from the Noble Research Institute to the University of Wisconsin-Madison to continue with the work and is now a CoI on the project.		
COI Name (Institution):	Chin, Sabrina Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	80NSSC22K0029		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: This project is a continuation with Principal Investigator (PI) Dr. Simon Gilroy at the University of Wisconsin of "The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth," grant 80NSSC18K1462. The previous PI was Dr. Elison Blancaflor, Noble Research Institute, from July 2018-March 2021. Plants will be a major component of advanced life support systems to enable NASA to realize its vision of long duration space exploration that will take humans to Mars and beyond. However, plants can only be effectively used for such systems if we have a deeper mechanistic understanding about how their development is regulated by the spaceflight environment. Recent "omics" studies have shown that plants reprogram their genetic circuitry when grown in the near weightless conditions of space. Genes involved in cell wall remodeling and oxidative stress are major gene categories that are differentially regulated by microgravity. Further, it has been shown that microgravity in space and simulated microgravity using clinostats or random positioning machines elicit roughly similar root skewing responses in the model plant Arabidopsis thaliana. Root skewing was more pronounced in mutants to a root-expressed vegetative actin isoform. Consistent with the enhanced root skewing of actin mutants in microgravity is the observation that other plant species such as maize (Zea mays L.) have stronger root directional growth responses on a clinostat when chemicals such as latrunculin B disrupt their actin cytoskeleton. Based on these observations, we will test the hypothesis that the actin cytoskeleton is a key player in dictating root growth directionality in microgravity and simulated microgravity by regulating the process of autotropic organ straightening. Moreover, we hypothesize that a set of cell wall cross-linking (i.e., class III peroxidases) and putative actin-regulatory proteins (i.e., HLB1 and MIN7/BEN1) mediate root hair elongation and directional root growth, respectively, under microgravity and simulated microgravity. Our hypotheses will be addressed through three specific aims including the analysis of cell wall composition/distribution in wild-type Arabidopsis roots, and detailed characterization of selected Arabidopsis cell wall and actin mutants subjected to microgravity simulation on a slow rotating clinostat and random position machine (aim 1 and 3). We will also use microgravity simulators to address the hypothesis that the plant hormone brassinolide regulates directional root growth in maize through its modulation of actin organization and dynamics (aim 2). Our research plan builds on successes of previous NASA-funded ground and spaceflight results, and outlines a comprehensive set of experiments including cell wall immunocytochemistry, computer-based root growth analyses, actin quantification, and genetics. The proposed work aligns with Plant Biology (PB) Element of the Space Biology Science Plan. Results obtained from plant experiments on microgravity simulators will address guiding questions linked to the decadal survey recommendations including the impact of gravity on plant growth, development and metabolism (PB-1), and molecular mechanisms underlying how plants sense and react to gravity (PB-5). We expect that the proposed research will provide us with basic knowledge that will guide the development of plant cultivars better adapted to the microgravity environment of spaceflight, an outcome that will prove beneficial for the design of plant habitats on future space colonies and for addressing great agricultural challenges on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to gain a better mechanistic understanding of how plant roots function at the cellular and molecular levels. A primary focus of the proposed work is to ask how the microgravity environment modulates the directionality of root growth. Because roots are essential for the uptake of water and nutrients, knowledge gained from successful implementation of the proposed work will be crucial in guiding pre-breeding efforts aimed toward the development of crops with improved productivity through better and more efficient root systems. Thus, agriculture on Earth is one beneficial outcome of this project.
	This grant is a continuation of NASA grant 80NSSC18K1462, which was transferred from the Noble Research Institute to the University of Wisconsin. Research accomplishments since the grant was transferred to Wisconsin are outlined below. Remaining experiments are related to aims 1 and 3. Experiments in aim 2 were completed at the Noble Research Institute. Specific Aim 1. Immunocytochemistry of selected cell wall components in roots of Arabidopsis thaliana seedlings grown on a slowly rotating clinostat and random positioning machine (RPM). Status: in progress. Seedlings from experiments growing plants on 3d clinostats at the Microgravity Simulation Support Facility (MSSF) at NASA Kennedy Space Center are currently stored in Phosphate Buffered Saline (PBS) or embedded in London Resin (LR) White Resin, awaiting sectioning and immunolabeling. However, for this reporting period, we have prioritized completing the glycomics and immunocytochemistry of seedlings from the previous APEX-03 Arabidopsis flight experiment. This data will then guide our work on immunocytochemistry of the seedlings grown on microgravity analogs. Glycomics uses a panel of >100 antibodies against wall components to define wall composition. The data we have collected on glycomics and immunocytochemistry from APEX-03 showed differences in the abundance of non-cellulosic cell wall components between ground- and space-grown seedlings. Glycomics revealed that spaceflight induced a general increase in non-cellulosic wall components based on high binding intensity of monoclonal antibodies to wall components from the space-grown seedlings. Immunocytochemistry of 11-day-old seedlings roots validated results from the glycomics readout. The cell wall components that changed in spaceflight will be used for immunocytochemistry of seedlings grown on a 3-D clinostat and RPM to determine if these microgravity analogs mirror spaceflight conditions. Specific Aim 2. Characterize autotropic root straightening in maize roots in response to brassinolide (BL) - and latrunculin B (LatB) - induced actin disruption. Status: completed at the Noble Research Institute. Specific Aim 3. Evaluate root growth of selected Arabidopsis thaliana actin single and double mutants, and peroxidase mutants on microgravity simulators. Status: in progress. During this reporting period, progress was made in characterizing actin cytoskeletal-related mutants in Arabidopsis thaliana. One significant accomplishment is the characterization of a mutant that is resistant to the growth inhibitory effects of the actin-disrupting compound LatB. The rationale for this aim is based on the observation that LatB enhances root gravitropism by inhibiting autotropic root straightening. Therefore, we expect that mutants that are hypersensitive or resistant to LatB could shed light into actin-mediated root development in microgravity. Progress to date is primarily on mutants that are hypersensitive to LatB (i.e., the hlb mutants). Previous publications described the characterization of these hlb mutants and identification of the disrupted genes that confer LatB hypersensitivity. Proteins encoded by the HLB genes include a trans-Golgi Network (TGN)-localized protein involved in exocytosis and endocytosis, a BEACH-domain containing protein called SPIRRIG (SPI) that functions in actin-mediated root hair growth, and the actin-nucleating class II formin 20.

Task Progress:	For this reporting period, we completed the characterization of one LatB-resistant mutant (hereafter referred to as LBR1). In the absence of LatB, LBR1 and wild type have similar root growth phenotypes. However, when three-day-old wild type and LBR1 seedlings were transferred to Murashige-Skoog (MS) plates supplemented with a range of LatB concentrations, it was found that LBR1 primary root length was longer than that of wild type. Robust primary root growth in LBR1 was observed even at LatB concentrations that severely inhibited that of wild type. The ability of LBR1 to tolerate the inhibitory effects of LatB was also observed in other organs and cell types. For instance, hypocotyls of dark-grown LBR1 mutants incubated in LatB-supplemented MS plates were significantly longer than those of wild type. Furthermore, LBR1 root hairs still emerged at LatB concentrations that typically blocked root hair initiation in wild type. The ability of LBR1 to tolerate the inhibitory effects of LatB led to the hypothesis that the LBR1 actin cytoskeleton remains intact even when exposed to LatB. To test this hypothesis, stable lines of LBR1 expressing the filamentous-actin (F-actin) reporter, UBQ10:GFP-ABD2-GFP were generated. F-actin in wild type and LBR1 primary roots was examined two days after transferring seedlings to 100 nM LatB. At 100 nM LatB, F-actin networks in LBR1 primary roots remained dense, while wild type exhibited fragmented F-actin. The data indicate that the actin cytoskeleton of LBR1 is more resistant than that of wild type to LatB, and therefore consistent with the ability of LBR1 seedlings to grow at LatB concentrations that severely inhibit the wild type. A cross between LBR1 and wild type resulted in progeny that exhibited resistance to LatB, which indicated that LBR1 is a dominant mutant. To determine the identity of the LBR1 gene, whole genome sequencing comparison between lbr1 mutant and wild type plants was conducted. This revealed that LBR1 had a mutation in the AT5G09810 gene which encodes ACTIN7 (ACT7), which is one of three vegetative actin isoforms in Arabidopsis. To confirm if LBR1 is ACT7, LBR1 plants were transformed with wild type ACT7 under the control of the ACT7 promoter (pACT7:ACT7). Furthermore, wild type plants were transformed with the mutated LBR1 under control of the ACT7 promoter (pACT7:LBR1). Results showed that LBR1 mutants expressing pACT7:ACT7 exhibited wild type sensitivity to LatB, while wild type plants expressing pACT7:LBR1 became LatB resistant. The data confirm that LBR1 is ACT7. During this reporting period, we also completed genetic interaction studies between SPI/HLB2 and components of the ARP2/3-W/SCR pathway. Mutations in SPI/HLB2 lead to short root hairs and hypersensitivity to LatB. SPI/HLB2 mutants also have mild trichome shape defects. Because SPI/HLB2 mutants partly resemble mutants to the actin-nucleating WAVE/SCAR-ARP2/3 complex, we hypothesized that SPI/HLB2 might function in overlapping genetic pathways as WAVE/SCAR-ARP2/3 in actin-mediated plant cell development. We tested this hypothesis by generating double mutants between SPI/HLB2 and WAVE/SCAR-ARP2/3, and analyzed their root hairs, trichomes, and primary root sensitivity to LatB. For LatB sensitivity assays, we found that primary roots of arp2, arp3, arpc5, and brk1 exhibited some hypersensitivity to LatB. However, the response of these mutants to LatB was not as severe as spi and leaned more toward wild type sensitivity. By contrast, the hypersensitivity of primary roots of spi arp2, spi arp3, spi arpc5 and spi brk1 double mutants to LatB was more similar to spi suggesting genetic interactions occurs between SPI/HLB2 and W/SCR-ARP2/3. Genetic interaction between SPI/HLB2 and W/SRC-ARP2/3 was also observed in trichomes. Consistent with previous reports, arp2, arp3, arpc5 and brk1 had more severe trichome shape defects than that of wild type and spi, while those of spi were mild when viewed with a scanning electron microscope. Arp2, arp3, arpc5, and brk1 trichomes had distinctively shorter branches than that of wild type and spi/hlb2. Whereas trichomes of spi were wavy and crooked, they were not short as those of arp2, arp3, arpc5, and brk1. We found that trichomes of all the double mutants resembled those of spi/hlb2, which provided further validation that SPI functions in overlapping genetic pathways as WAVE/SCAR-ARP2/3. We also completed experiments on imaging the actin cytoskeleton in developing trichomes of spi/hlb2 mutants. Data showed that actin organization in spi/hlb2 trichomes was similar to that of wild type, which was consistent with the mild trichome defects of spi/hlb2.

Bibliography Type:	Description: (Last Updated: 02/02/2026)
Books/Book Chapters	Chin S, Blancaflor EB. "Plant gravitropism: From mechanistic insights into plant function on Earth to plants colonizing other worlds." in "Plant Gravitropism Methods and Protocols." Ed. E.B. Blancaflor. New York, NY: Humana Press, 2022. p. 1-41. https://doi.org/10.1007/978-1-0716-1677-2_1 , Jan-2022
Books/Book Chapters	Chai C, Chin S, Blancaflor EB. "Imaging the cytoskeleton in living plant roots." in "Cytoskeleton Methods and Protocols." Ed. R.H. Gavin. New York, NY: Humana Press, 2022. p 139-148. http://doi.org/10.1007/978-1-0716-1661-1_6 , Jan-2022
Books/Book Chapters	Avci U, Nakashima J. "A flat embedding method to orient gravistimulated root samples for sectioning." in "Plant Gravitropism Methods and Protocols." Ed. E.B. Blancaflor. New York, NY: Humana Press. 2022. p. 153-164. https://doi.org/10.1007/978-1-0716-1677-2_11 , Jan-2022